

31338 Network Servers

Lab 10a: Basic web server configuration

Aims

1. To configure a Linux machine to act as a web server using Apache

Task 1: Verify installation and start the web server

The Apache web server should be installed, but it is probably not running. Check the status with:

```
systemctl status httpd
```

Also check that `mod_ssl` is installed (we need this for the next lab): `rpm -qa | grep mod_ssl`

If either of them are not installed, use `dnf` to install as previously.

Check to see what files were installed (hint: `rpm -qs httpd` and `rpm -qs httpd-filesystem`).
The main web pages are installed in `/var/www` and the configuration files are in `/etc/httpd`.

Use `systemctl` to start your `httpd` daemon now, and also use `systemctl` to enable your `httpd` daemon at boot time.

Test the server by opening Firefox and trying **`http://localhost`**

You should see the default Apache startup page.

In the next exercise we will modify the configuration.

Task 2: Basic web server functionality

First, make sure your DNS is working for your domain (e.g. `it.netserv.edu.au`). If you didn't previously, you may also like to add an entry to the DNS for `www.it.netserv.edu.au` whose IP address is that of your virtual machine. As noted in previous labs, make sure that there is consistency in what IP address appears in your zone file, and the IP address currently on Ethernet interface `ens37`. Also, make sure that the top "nameserver" line in the `/etc/resolv.conf` points to your nameserver. If your DNS needs fixing, refer to the earlier DNS lab. You need DNS working for the web server configuration below.

Edit the `/etc/httpd/conf/httpd.conf` file. To start with, you should modify *ServerName*, and note down the *DocumentRoot* directory. *DocumentRoot* is where the HTML documents for your web site should be placed. Restart your web server: `systemctl restart httpd`

You can test your web server by using the <http://www.it.netserv.edu.au> (or whatever you set up) in the browser.

You can create a file named `index.html` in your *DocumentRoot* directory. By default your *DocumentRoot* is `/var/www/html`, unless you changed it when you edited the `httpd.conf` file.

Common problems:

- When you attempt to start/restart the `httpd` daemon, if you get a message about unable to bind to port 80, this is probably because you need to be root to start the web server!
- Cannot view my `index.html`? eg: 404 error or 403 error? Check that the file exists in the correct *DocumentRoot* and/or the file has public read permission and directory has have public read/execute permissions.
Try executing `ps -ef | grep httpd`. Note that the web service runs under the userid `apache`. Hence all files in the `/var/www` directory need to be able to be read by userid `apache`.
- Getting a FAILED error on startup, or an error 500 from the browser. This is usually a configuration error. Use `apachectl -t` to check your config.